

ADVANCED STATISTICAL MECHANICS

Problem 1 (3.14)

Consider a system of N classical particles with mass m moving in a cubic box with volume $V = L^3$. The particles interact via a short-ranged pair potential $u(r_{ij})$ and each particle interacts with each wall with a short-ranged interaction $u_{\text{wall}}(z)$, where z is the perpendicular distance of a particle from the wall.

Write down the Lagrangian for this model and use a Legendre transformation to determine the Hamiltonian H .

(a) Show that the quantity

$$P = - \left(\frac{\partial H}{\partial V} \right) = - \frac{1}{3L^2} \left(\frac{\partial H}{\partial L} \right)$$

can be identified as the instantaneous pressure, i.e., the force per unit area on the walls.

(b) Reconstruct the Lagrangian in terms of the relative locations of the particles inside the box $r_i = Ls_i$, where the vectors s_i lie inside a unit cube. Use a Legendre transformation to determine the Hamiltonian with this set of variables.

(c) Recalculate the pressure using the second version of the Hamiltonian. Show that the pressure now includes three contributions:

- (1) a contribution proportional to the kinetic energy per particle,
- (2) a contribution related to the forces between pairs of particles, and
- (3) a contribution related to the force on the wall.

Show that in the thermodynamic limit the third contribution is negligible compared with the other two. Interpret contributions 1 and 2 and compare with the virial equation of state (3.7.15).

Let's write the variables of interest in this question :

We have N identical particles of mass m in cube of side length L .

• Position of particle i : $\vec{r}_i = (r_{ix}, r_{iy}, r_{iz})$

• Velocity : $\vec{r}_i'$

• Pair potential between particles i and j : $u(r_{ij})$ where $r_{ij} = |\vec{r}_i - \vec{r}_j|$

• Wall potential between particle i and all six walls : $\sum_{\alpha=\{x,y,z\}} u_{\text{wall}}(r_{i\alpha}) + u_{\text{wall}}(L - r_{i\alpha}) \Rightarrow \text{Potential from one coordinate}$

• Momentum : $\vec{p}_i'$

We are ready to write the Lagrangian of the system.

$$\begin{aligned} \mathcal{L} &= \text{Kinetic Energy} - \text{Potential Energy} \\ &= \sum_{i=1}^N \frac{m \vec{r}_i'^2}{2} - \sum_{i < j} u(r_{ij}) - \sum_{i=1}^N \sum_{\alpha=\{x,y,z\}} u_{\text{wall}}(r_{i\alpha}) + u_{\text{wall}}(L - r_{i\alpha}) \end{aligned}$$

To find the Hamiltonian, we perform the Legendre transformation:

$$H(\vec{r}, \vec{p}) = \sum_i \vec{p}_i \cdot \vec{\dot{r}}_i' - \mathcal{L}(\vec{r}, \vec{\dot{r}}')$$

$$= \sum_i m \vec{\dot{r}}_i'^2 - (K - V)$$

$$= 2K - K + V = K + V \quad \text{where } K \text{ is the Kinetic Energy}$$

& V is the Potential Energy

$$= \sum_{i=1}^N \underbrace{\frac{\vec{p}_i^2}{2m}} + \sum_{i < j} U(r_{ij}^0) + \sum_{i=1}^N \sum_{\alpha=\{x,y,z\}} U_{\text{wall}}(r_{i\alpha}) + U_{\text{wall}}(L - r_{i\alpha})$$

$$\vec{p}_i = \frac{\partial \mathcal{L}}{\partial \vec{\dot{r}}_i} = m \vec{\dot{r}}_i'$$

a) Goal: Show that $\frac{\partial H}{\partial V} = -P$.

Solution

Since our volume is encoded in our side length L . We will use the chain rule.

$$V = L^3 \Rightarrow \frac{\partial V}{\partial L} = 3L^2$$

$$\frac{\partial H}{\partial V} = \frac{\partial H}{\partial L} \cdot \frac{\partial L}{\partial V} = \frac{1}{3L^2} \cdot \underbrace{\sum_i \sum_{\alpha} U'_{\text{wall}}(L - r_{i\alpha})}_{\substack{\text{only wall terms of} \\ \text{right, back \& top} \\ \text{contribute}}} = \frac{1}{3L^2} \cdot \underbrace{(F_x + F_y + F_z)}_{\substack{\text{Force} \\ \uparrow \uparrow \uparrow \\ \text{area}}} = \underbrace{-P}_{\substack{\text{accounting} \\ \text{for direction} \\ \text{of } P}}$$



b) Goal: Rescale the Lagrangian & Hamiltonian for a unit cube

Solution

Let $\vec{s}_i = \vec{r}_i / L$ be the new scaled positions for our particles. Then, the Lagrangian is:

$$\begin{aligned} \mathcal{L} &= \sum_{i=1}^N \frac{m \vec{\dot{r}}_i'^2}{2} - \sum_{i < j} U(r_{ij}^0) - \sum_{i=1}^N \sum_{\alpha=\{x,y,z\}} U_{\text{wall}}(r_{i\alpha}) + U_{\text{wall}}(L - r_{i\alpha}) \\ &= \sum_{i=1}^N \frac{m L^2 \vec{\dot{s}}_i'^2}{2} - \sum_{i < j} U(L s_{ij}^0) - \sum_{i=1}^N \sum_{\alpha=\{x,y,z\}} U_{\text{wall}}(L s_{i\alpha}) + U_{\text{wall}}(L - L s_{i\alpha}) \end{aligned}$$

$$H = \sum_{i=1}^N \underbrace{\frac{\tilde{p}_i^2}{2mL^2}} - \sum_{i < j} U(L s_{ij}^0) - \sum_{i=1}^N \sum_{\alpha=\{x,y,z\}} U_{\text{wall}}(L s_{i\alpha}) + U_{\text{wall}}(L - L s_{i\alpha})$$

$$\tilde{p}_i = \frac{\partial \mathcal{L}}{\partial \vec{\dot{s}}_i'} = mL^2 \vec{\dot{s}}_i' = L \vec{p}_i \quad (1)$$

c) Goal = Recalculate P for rescaled H .

Solution

$$\begin{aligned}
 P &= -\frac{1}{3L^2} \frac{\partial H}{\partial L} = -\frac{1}{3L^2} \partial_L \left[\sum_{i=1}^N \frac{\tilde{p}_i^2}{2mL^2} - \sum_{i<j} U(Ls_{ij}) - \sum_{i=1}^N \sum_{q=\{x,y,z\}} U_{\text{wall}}(Ls_{iq}) + U_{\text{wall}}(L-Ls_{iq}) \right] \\
 &= \frac{1}{3L^2} \sum_i \frac{\tilde{p}_i^2}{mL^3} \quad \rightarrow \text{kinetic Energy contribution} \\
 &\quad - \frac{1}{3L^2} \sum_{i<j} s_{ij} U'(Ls_{ij}) \quad \rightarrow \text{interaction between particles} \\
 &\quad - \frac{1}{3L^2} \sum_i \sum_q s_{iq} U'_{\text{wall}}(Ls_{iq}) + (1-s_{iq}) U'_{\text{wall}}(L-Ls_{iq}) \quad \rightarrow \text{interaction between particles \& walls.}
 \end{aligned}$$

Converting back to the original coordinates: $\vec{r}_i = L\vec{s}_i$ & $\vec{p}_i = \frac{\tilde{p}_i}{L}$ (1)

$$\begin{aligned}
 P &= \frac{2}{3L^3} \sum_i \frac{\tilde{p}_i^2}{2m} \\
 &\quad - \frac{1}{3L^3} \sum_{i<j} r_{ij} U'(r_{ij}) \\
 &\quad - \frac{1}{3L^3} \sum_i \sum_q r_{iq} U'_{\text{wall}}(r_{iq}) + \underline{(L-r_{iq})} U'_{\text{wall}}(L-r_{iq}) \quad \sim O\left(\frac{L}{L^3}\right) \\
 &\quad \rightarrow O(V^{-2/3}) \\
 &\quad \rightarrow O(N^{-2/3}) \rightarrow 0 \quad \text{for } N \rightarrow \infty \text{ thermodynamic limit}
 \end{aligned}$$

$$P = \frac{2}{3V} N \langle U \rangle$$

$$- \frac{1}{3V} N \left\langle \sum_{i<j} U'(r_{ij}) r_{ij} \right\rangle \rightarrow \text{virial of a particle}$$

Replacing $\langle U \rangle = \frac{3}{2} kT$

$$P = \frac{NkT}{V} - \frac{1}{3V} N \left\langle \sum_{i<j} U'(r_{ij}) r_{ij} \right\rangle$$

↑ Virial State Equation with dimension 3